

Scale-free dynamics of visual inference

Jude Luca Metcalf

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Prof. Pulin Gong

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Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

Jude Luca Metcalf

August 4, 2026

Abstract

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Acknowledgements

Contents

Abstract iii

Acknowledgements iv

Introduction 1

1 Survey of the Literature 2
- 1.1 Stochastic modelling . . . 2
  - 1.1.1 Stochastic calculus . . . 2
  - 1.1.2 Anomalous diffusion . . . 3
  - 1.1.3 Lévy stable motion . . . 4
- 1.2 Abstract computation . . . 5
  - 1.2.1 Active inference . . . 5
  - 1.2.2 Neural sampling . . . 5
  - 1.2.3 Markov chain monte carlo methods . . . 7
- 1.3 Behavioural data . . . 8
  - 1.3.1 Neural frequency bands . . . 8
  - 1.3.2 Gaze trajectories . . . 8

Bibliography 10

Introduction

The brain is among the most widely studied subjects of research, and yet many neural behaviours remain elusive to scientists. In recent years, methods adapted from statistical physics, network science, and thermodynamics have been found to explain and provide understanding for computation in the brain. A large part of the brain is devoted to vision, the visual cortex containing between 4-6 billion of the 30 billion neurons of the neocortex, up to 5 times the size of other primates [1, 2]. Vision is thus an important neural process, and provides one of the first gates of attention to the human brain.

Many approaches to understanding patterns of human gaze have been described. Importantly, there has been success in training large neural networks to predict gaze trajectories [3–8] and saliency maps from an image for human observers [9–14]. Such approaches give often quite accurate results, but they lack the rich understanding of a full model of the network of the brain itself. Theories of biological attention and cognition, such as those of the "Bayesian brain", built on active sensing and inference provide a more motivated framework to understand the brain as maintaining a rational causal mental worldview [14–17]. This Bayesian model of the brain has been a fruitful direction to understand the most basic level of neural cognition as sampling through population activity patterns [18–21]. Although it has been a debated framework (see [22]), the Bayesian brain has provided a strong window into biological cognition. Geoffrey Hinton and colleagues invented the Helmholtz machine, performing explicit Bayesian inference on perceptive inputs, and updating predictions on the external world, minimising the free energy as a loss function of a neural network. Such a formulation inspired pioneering work on neural coding [20, 23], Karl Friston's highly influential Free Energy Principle in theoretical neuroscience [24, 25], and the resulting empirical models and well-verified predictions about the human sensory-motor system [26–28].

Chapter 1: Survey of the Literature

This thesis aims to further our understanding of the details of biological neural computation through understanding the dynamics of visual searching. There is growing success in turning methods from statistical physics towards the area of computational neuroscience, and major contributions to this project are described here. In particular, the rich scale-free activity of neural circuits inspires similar analysis to the dynamics of gaze trajectories, alongside the information-theoretic pressure of the visual system to process data near-optimally. We aim to understand these behaviours as an efficient method of gathering information. In Section 1.1 we introduce the framework of stochastic modelling, Lévy motion, and the associated anomalous superdiffusion which informs the dynamics of the FNS sampler. Section 1.2 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts. Section 1.3 provides the background for much of the behavioural data we present and model in this thesis, and the previous methods used to understand and model the complex dynamics found in gaze trajectories especially.

1.1 Stochastic modelling

1.1.1 Stochastic calculus

Stochastic calculus has been fundamental in understanding how high dimensional complex systems reduce to a small number of variables of interest. When studying complex systems in physics, economics, geology, climate science, machine learning, and neuroscience, it is often the most natural, tractable, and accurate modelling choice when understanding a system of interest. The important concept here is that of the introduction of some central limit theorem, where the sum of very many small interactions which are understood microscopically can produce certain well-defined stochastic patterns on macroscopic scales of interest. The ubiquitous stochastic process is a Wiener process W_t, equivalently called Brownian Motion. A Wiener process is defined by the stationary and independent Gaussian increments across all non-overlapping time intervals. The Wiener process is a very simple and analytically elegant model of the coarse-grained effect of very many small finite-variance interacting influences. A Wiener process is defined only by its timescale, with a variance over trajectories after time t of t itself. Wiener processes yield time series which are almost surely continuous but nowhere differentiable. The Wiener process is understood as a model of the long-time dynamics of a variable effected by very many small uncorrelated finite variance influences [29]. Classical examples of processes modelled by the Wiener process are: the movement of a macroscopic particle in a non-turbulent fluid [29], the gravitational influence of an orbiting thermal collection of stars on supermassive black holes [30, 31], and (arguably) the movement of low-volatility stock prices [32]. There are well studied mechanisms through which complex real-world stochastic processes diverge from the Wiener process. Of interest to us is the case where the microscopic effects become correlated with heavy-tailed statistics. Correlated steps, long-time memory effects, and microscopic dynamics can result in a stochastic process which does not limit to the Wiener process [33, 34]. Of interest to us is the fractional and superdiffusive Lévy flight, which is a result of spatial non-locality in an interacting network. For instance, previous studies have demonstrated a neural circuit network at criticality resulting in diverging correlation length of activity, and the centre of mass of neural firing is subject to the large jumps. Such large jumps in a stochastic time series are modelled according to

Lévy stable motion, where the increments are drawn independently from a heavy-tailed distribution. This section describes the analytics and behaviour of such Lévy processes for our current study.

1.1.2 Anomalous diffusion

A key motivation for the development for stochastic modelling was when studying processes of diffusion. In 1885, the German physiologist Adolf Fick was the first to describe the universality of normal diffusion [35]. Fick records the diffusion equation, which is identical to the heat equation applied to a probability distribution, as

$$\frac{\partial p(x,t)}{\partial t} = D\frac{\partial^2 p(x,t)}{\partial x^2} \tag{1.1}$$

where D is a scaling constant. Fick shows that this gives a mean squared displacement (MSD) $\langle x(t)^2\rangle = 2Dt$. This MSD linear in time defines normal diffusion. Such normal diffusion is found in many simple random processes, such as the paradigm case of the Brownian motion of dust in a fluid studied by Albert Einstein. Both Albert Einstein and the British statistician Karl Pearson independently developed the continuous time random walk (CTRW) with finite-variance step size and step time distributions, robustly producing this normal diffusion. Increasingly, physical systems have been which diffuse nonlinearly in time, a situation known as anomalous diffusion, with an MSD scaling as:

$$\langle x(t)^2\rangle \propto t^\beta, \quad \beta \neq 1 \tag{1.2}$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \beta < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \beta < 2$ where diffusion scales faster than normal. The case of $\beta = 2$ is known as ballistic diffusion, as it is found in the deterministic case of constant velocity $x = vt$. Anomalous subdiffusion has been shown to result from heavy-tailed wait-times between interactions [35], such as in trapped fluid flow [36], controlled diffusion through cell membranes [35], and protein mixing [37].

Anomalous superdiffusion

More importantly to this thesis, superdiffusion has been studied in the contexts of complex exploratory biological systems. Superdiffusion has a strong link to the problem of a blind search for scarce resources. The Lévy foraging hypothesis posits that a superdiffusive CTRW (known as a Lévy walk) is an optimal strategy when searching for sparse and randomly distributed resources. This result was first reported in a letter to the journal Nature in 1999 by Viswanathan et al. [38]. In this paper a Lévy walk with superdiffusion $\langle x(t)^2\rangle \propto \frac{t^2}{\ln(t)}$ on the transition to ballistic diffusion was found to be optimal in a search for sparse resources. The optimality of the superdiffusive Lévy walk has been reproduced in diverse functional search models [39–41], in a population of searching agent [42], and in the empirical search paths of biological organisms, from the flights of drosophilia flies [41, 43], albatross' searches [44, 45], and strongly across many marine predators [46, 47]. Increasingly there are studies which describe biological limitations of the Lévy foraging hypothesis, however the Lévy computational advantage in a diverse set of resource landscapes without the need for fine-tuning remains strong [39, 40]. This marks a split in the treatment of Lévy walks, with groups studying them as informative baseline functional models of abstract search [38, 39, 42, 48], and others led by authors such as Graham Pyke presenting this as a misrepresentation of true intelligent organisms [49–51], see Reynolds

(2015) for an in-depth discussion of a richer view of biological Lévy walk superdiffusion [52].

1.1.3 Lévy stable motion

A general superdiffusive process which is highly analytically tractable, not least because it is Markovian, is the *Lévy flight*. The nomenclature in the literature is often unclear when distinguishing stochastic processes of a heavy-tailed nature. Here, the Lévy flight L_T^α is a stochastic process defined through it's increments being drawn from a heavy tailed symmetric stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution). $\mathcal{S}\alpha\mathcal{S}$ distributions describe the limit of summation of i.i.d infinite-variance variables,as a result of the generalised central limit theorem. The α value defining a particular $\mathcal{S}\alpha\mathcal{S}$ distribution is a result of the asymptotic power-law tail of the density function, with $p(x) \propto |x|^{-1-\alpha}$. Thus, the Lévy flight L_t^α after time t is defined as follows, to outline the fractional scaling properties and alpha-stable statistics

$$L_t \overset{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \tag{1.3}$$

where the parameter α defines the heaviness of the tails of resulting statistics, and where $\overset{d}{=}$ denotes equal in distribution. In the limit $\alpha \to 2$, the Lévy flight reduces to a scaled Brownian Motion $L_t^{\alpha\to 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges and has mean 0, whereas if $0 < \alpha \leq 1$, then both the variance and mean diverge. In analysing the distribution of a Lévy flight, one must employ fractional calculus in the form of the fractional Fokker-Planck equation. The details of such an analysis are beyond this thesis, however the fractional Fokker-Planck equation is the tool used to understand the superdiffusion, nonlocality, and scale-free behaviour of trajectories of Lévy flights (see the appendices to [53]). Many of the simple analytic properties of the Wiener process are no longer true of a Lévy flight. Importantly, trajectories of a Lévy flight are almost surely no longer continuous. Instead, such trajectories are called *càdlàg* (continue à droite, limite á gauche) with respect to time, which describes functions which are right continuous with well-defined left limits [54]. Thus, we say that since all such trajectories produced from a Lévy flight are càdlàg, the Lévy flight itself is called a càdlàg process. Càdlàg processes have been widely studied in the context of mathematics, economics, and physics, and their properties are well understood [55, 56]. It has been shown that Lévy flights are semimartingales, and are well interpreted in integration [54]. Importantly, it follows from the theorem of the Lévy-Itô decomposition that a Lévy flight can be decomposed as the sum of two processes [57]:

1. A compound Poisson process of large jumps with $|\Delta x| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number of such jumps in any finite time period.

2. A square-integrable martingale (mean 0) jump process with $|\Delta x| < 1$. This process is thus finite-variance, and is thus subject to the central limit theorem, meaning that under temporal coarse graining it approaches a Wiener process.

This division is useful in analysing the process in the context of nonlocal stochastic exploration in Lévy flights, which is modelled by the compound Poisson process of large jumps $|\Delta x| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $|\Delta x| < 1$.

1.2 Abstract computation

Probabilistic sampling is a powerful and flexible paradigm of general computation. Sampling methods represent and evaluate probability distributions through the production of stochastic samples samples. Uncertain, high-dimensional, and ambiguous inference problems require efficient and accurate characterisations of target distributions. Sampling methods draw from a target distribution such that the sampled ensemble distribution reflects the target itself. This paradigm has become widely utilised in machine learning and statistics, where generative models learn patterns from large multivariate distributions inferred from observational data [58, 59]. This framework has gained significant traction as a basic scheme to describe biological computation. The position that the brain itself acts as a Bayesian generative model has become that interprets through continuous sampling mechanisms has become well supported.

1.2.1 Active inference

The dominant framework in modern computational neuroscience is that of the Bayesian brain. Karl Friston, the most cited neuroscientist [60], developed this framework, and shifted the view of the perceptual loop of the brain. Friston's theory of active inference described perception through inferring and generating top-down sensory predictions of the world, and carrying back the bottom-up errors from that perception. This development brought a strong shift in the literature, "marking a watershed between 20th-century thinking about the brain as a glorious stimulus–response link and more constructivist 21st century perspectives that emphasized an active sampling of the sensory world" ([61] pg. 1019). The active inference model of an updating internal mental state filled the explanatory gap in the understanding of the functional role of many key feedback mechanisms in the brain [27, 28], and went on to yield successful predictions of how the brain optimally deciphers noisy multimodal sensory inputs [26], as well as provide rigorous frameworks to base probabilistic computation in the brain [62]. There has recently been opposition to the general framework of the Bayesian brain and active inference [22]. Much of the arguments against such a framework arise from the philosophy of science, where active inference in perception may be supported post-hoc through "just so" definitions of priors [22, 63]. It is therefore important to understand the Bayesian brain hypothesis as a normative framework from which to build falsifiable, understandable, and biologically plausible theories of neural computation.

Visual search

This thesis presents visual search tasks as such a paradigm where a Bayesian model is well motivated. The visual system is evolved to actively and efficiently process noisy information and evidence from the external world, and plan future search paths to gather further noisy evidence [14, 64–66]. In particular, search tasks require sophisticated integration of the cost and expected information gain of decisions, which invites fruitful decision analysis [67]. Empirical experiments show that eye movements show good agreement with greedy local information maximising fixations, given well-motivated visual sensitivity based on human physiology [65, 66, 68].

1.2.2 Neural sampling

Another useful paradigm for the basic functional role of neural activity is the neural sampling hypothesis. The framework of neural sampling posits that the variability of cortical neural responses is an important functional feature of biological computation.

Otherwise regarded as noise, stochastic neural activity patterns are thus presented as samples drawn from a posterior distribution over possible sensory interpretations [18, 19]. The neural sampling hypothesis has been shown to perform and predict network-level empirical patterns of neural activity in the V1, and recently by Terada and Toyoizumi in biologically realistic cortical networks, better than alternative theories [18, 19, 69, 70]. Perceptual uncertainty is then encoded by the variability of these cortical responses across time and across multiple trials, with higher uncertainty reflected in wider response distributions [18, 19, 58]. Neural sampling thus provides an interpretation for the way the brain navigates uncertainty at the fundamental scale of neural computation.

Studies into neural sampling show that variability is not strictly a result of explicit stochastic noise, as traditionally believed. Instead, recent work demonstrates that strongly recurrent, balanced inhibitory-excitatory deterministic neural networks can generate the high-dimensional variability which stochastic models are built to capture [19, 70]. Terada and Toyoizumi link this sampling of a posterior distribution to Bayesian learning through training a recurrent network to sample static features and dynamic trajectories, generalising learned priors to novel, uncertain environments through biologically plausible learning [70]. Further biologically plausible implementations of such Bayesian sampling are energy-based models (EBMs) which suggest that fast-spiking interneurons can estimate the local partition function to enable fast, localised sampling dynamics [71]. The sampling hypothesis has thus given a convincing theory of biological neural computation, inspired by the Bayesian brain framework (as described above in 1.2.1), with testable and well-supported evidence that neural variability has an important functional role in computation [18, 69]. Traditional Gaussian and Poisson noise models representing these stochastic network dynamics inherently correspond to continuous struggle to capture the full dynamics of complex interconnected inhibitory-excitatory (IE) biological neuronal network [72–74]. Further study into the dynamics of these IE networks has thus motivated further functional models of neural activity.

Fractional neural sampling

In reality, neural activity has been found to exhibit structure across multiple spatial and temporal scales, often described by scale-free statistics [75–83]. This observation, together with the neural sampling hypothesis, motivates the design and theory of Fractional Neural Sampling (FNS) [53]. FNS describes the complex spatiotemporal dynamics of spiking neural circuits, where localised activity patterns propagate through superdiffusive Lévy flights. Such Lévy flight dynamics of the centre of mass of the firings with an underdamped Langevin equation, which we shall call the FNS equations of motion,

$$\begin{aligned} d\mathbf{x}_t &= \gamma b(\mathbf{x}_t)dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= \beta b(\mathbf{x}_t)dt \end{aligned} \tag{1.4}$$

where $\mathbf{x}_t$ is the vector of position, $\mathbf{p}_t$ is a generalised auxiliary variable analagous to a momentum of the same dimensionality, $\beta > 0$ is a parameter defining the strength of coupling between $\mathbf{p}_t$ and $\mathbf{x}_t$, $\gamma > 0$ is the noise strength parameter, $0 < \alpha \leq 2$ is the Lévy tail exponent, and dL_t^α is the Lévy stable noise drawn from a $\mathcal{S}\alpha\mathcal{S}$ distribution. The term $b(\mathbf{x}_t)$ captures a deterministic gradient term in the dynamics, defined such that the invariant distribution of $\mathbf{x}_t$ after long time t approaches the target distribution sampled, $\pi(\mathbf{x})$. The exact expression for $b(\mathbf{x})$ is derived in [53], and approaches the gradient of the logarithm of the density as $\alpha \to 2$, with the full form for any $0 < \alpha \leq 2$ given as

$$b(\mathbf{x}) = \frac{\mathcal{D}_{\mathbf{x}}^{\alpha-2}[\pi(\mathbf{x})\partial_{\mathbf{x}}\pi(\mathbf{x})]}{\pi(\mathbf{x})} \tag{1.5}$$

The FNS dynamics are called fractional, as the nonlocal dynamics of their distribution require fractional calculus to describe such spatial superdiffusion in Lévy flights [84]. The theory of FNS thus unifies the scale-free nonlocal dynamics of neural firing patterns with the neural sampling framework derived from the ideal Bayesian scheme of active inference. The critical advantage of FNS over other standard continuous random walks is its ability to efficiently sample complex, multimodal probability distributions with far-apart modes—likely a major feature of processed natural environments replete with multiple distinct salient features. While Brownian-motion-based samplers typically remain trapped in local modes due to high potential energy barriers, the heavy-tailed, large jumps inherent in FNS enable the neural sampler to adaptively and freely switch between multiple modes [85–88]. Furthermore, the generality of the FNS description permits the application of FNS equations of motion to gaze dynamics, as the motion of the eyes are driven by a IE neural circuit which is expected to keep key features of the FNS model invariant, such as superdiffusion, heavy-tailed statistics, and nonlocal interactions.

1.2.3 Markov chain monte carlo methods

This view of neural- and visual sampling of a target distribution motivates analogy to a foundational set of methods in machine learning. Markov Chain Monte Carlo (MCMC) methods have become the algorithmic basis for generating samples from target distributions in rigorous Bayesian inference in machine learning [59, 86, 89, 90]. MCMC operates by constructing a Markov chain, where the transition matrix is calculated from whose equilibrium distribution is the desired posterior distribution. In the context of neural computation, biological circuits are often modelled as implementing continuous-time variants of MCMC, most notably Langevin sampling and Hamiltonian Monte Carlo (HMC) [86, 91]. Langevin dynamics utilise local gradient information from the probability landscape, combined with Gaussian white noise, to explore the state space over time. Despite their ubiquity, traditional MCMC methods encounter severe theoretical and practical limitations when exploring unknown, heavy-tailed, and/or multimodal target distributions. Standard Langevin and HMC dynamics are driven by continuous Brownian and Hamiltonian motion respectively, and the traversal of low-probability regions, equivalent to dynamical energy barriers, becomes exponentially slow . Consequently, the mean exit time for the sampler to leave one mode and discover another grows exponentially with the modal separation distance [86]. To accelerate this sluggish exploration, algorithms often introduce auxiliary momentum variables. However, even with momentum-based acceleration, traversing deep energy valleys remains highly inefficient [85, 86, 91]. The fractional dynamics introduced in fractional sampling are one area of research into how fast mixing of samples in complex distributions can be achieved without the need for nontrivial parameter tuning. For instance, algorithms based on Hamiltonian Monte Carlo sampling and second-order Langevin sampling very similar to FNS dynamics have been briefly developed and explored [53, 87, 92]. Under FNS dynamics, the mean exit time between modes scales linearly rather than exponentially with separation, demonstrating a profound computational advantage for efficient biological Bayesian inference [53].

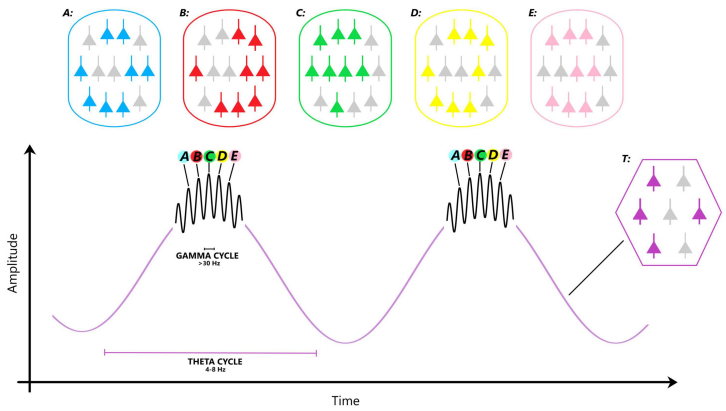


Figure 1.1: A schematic representation of theta-gamma coupling (TGC). TGC has it's foundations in the work of Lisman and Jensen in correctly ordering neural signals [105], and subsequently expanded and used to describe the dynamic mechanism the brain uses during exploration to modulate close attention upon individual centres of focus [106]. *Schematic adapted from figure 2 of M. Ursino and G. Pirazzini, 2024 [102]*

1.3 Behavioural data

1.3.1 Neural frequency bands

Throughout much of this thesis, links are drawn between timescales of interactions and dynamics in the FNS model and well-studied frequency bands of activity in the human brain. It is then valuable to outline the studied computational and functional interpretations of such frequency bands. Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data. Divisions are only approximate, but for this thesis these bands are: delta (0.1 - 4 Hz), theta (4 - 8 Hz), alpha (8 - 14 Hz), beta (14 - 30 Hz), and gamma (30 - 220 Hz) [93, 94]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to attention and search tasks of computation. Theta rhythms have been shown to modulate attention during auditory tasks [95, 96], visual search tasks [97–99], and in analysing the general attentional mechanism of the brain [17, 93, 94]. Intervals between theta peaks are finding stronger interpretation in the neuroscience literature as discretely sampling the environment for integration into the internal model of the brain. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [17, 95], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [97, 98]. Accordingly, the firing patterns which define the lowest level of information carried in the brain are regularly found to be gamma in scale ($\sim 30 - 100$ Hz) [100–104].

1.3.2 Gaze trajectories

Healthy human gaze trajectories consist of different dynamical regimes. The most studied property of the pattern of gaze trajectories are their division into periods of fixation, interspersed by near-instantaneous jumps between fixations, these jumps being known as "saccades" [107–110]. Importantly, this behaviour of local movements interspersed with large discontinuous jumps suggest modelling eye movements through a Lévy flight.

This method has a limited history in the literature, and fitting data to a simple phenomenological Lévy flight parametrised by the stable parameter α found limited use as a biomarker for ADHD [111]. Saccade metrics in general have been found to be valuable as non-invasive biomarkers for broader neural disorders, including schizophrenia, autism, ADHD, OCD, bipolar disorder, Parkinson's disease, and multiple sclerosis [112–116]. Periods of fixation have received less scholarly attention as a biomarker of cognition. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [117–120]. Examinations and literature reviews of these behaviours have often not considered their computational role in active sensing and efficient search.

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